

The subventricular zone concept: ready for therapeutic implications?

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See the article by Tejada Neyra and Neuberger et al. pp. 1517–1524.

For more than 10 years, with reaffirmation of the key role of radiation therapy (RT) in the postoperative management of most gliomas,¹ it has been a frustrating fact that no significant progress can be claimed in our specialty. Neither the “dose-escalation” nor the “dose-painting” concept, even when guided by “high-tech” imaging, permits significantly better local control that translates into longer survival. What we define as the CTV (clinical target volume), which should reflect the microscopic extension of glioma cells, is still a rather rough *isotropic* expansion of the GTV (gross tumor volume), usually 1.5–2 cm beyond.² Exploring new areas of potential progress in RT, several publications focused on glioblastoma (GBM) suggested that the subventricular zone (SVZ) could be the “niche” of glioma stem cells, which explains at least in part the inevitable recurrence of most gliomas: We would regularly miss the target, ignoring this key zone where the more radioresistant cells would hide.^{3,4}

In parallel, since 2016 and the new World Health Organization (WHO) classification, including key molecular alterations such as isocitrate dehydrogenase (IDH) mutation,⁵ IDH wild-type grade IV (primary GBM) was clearly separated from lower-grade gliomas (LGGs, WHO grades II and III), which are mainly IDH mutated, with this last subgroup representing roughly a third of the whole glioma population. Combining both the niche concept and a non-invasive way to visualize specifically the IDH-mutated glioma cells is logically one of the next steps to better understanding gliomagenesis.

In this area of “radiogenomics,” the study published in the current issue by the Heidelberg team evaluates the impact of tumor location on key molecular alterations in a large population of grade II to grade IV gliomas using a voxel-based lesion-symptom mapping (VLSM) tool.⁶ Hypothesizing that this region could be the niche of neural stem cells, the authors elegantly show that IDH-mutant gliomas are preferentially located in the frontal lobe, more precisely adjacent to the rostral extension of the lateral ventricles. Even if an association of brain tumor location with some molecular biomarkers was previously

suggested,^{7,8} including by one study using voxel-based analysis,⁷ authors extend here their study to all grades of gliomas, from LGG to GBM, adding to IDH mutation a large panel of most key molecular markers such as O⁶-methylguanine-DNA methyltransferase promoter methylation, epidermal growth factor receptor (EGFR) amplification, telomerase reverse transcriptase (TERT) gain for GBM, and 1p/19q codeletion status for LGG. They studied in depth and prior to surgery a large number of more than 350 patients, representing a nonselected “real-life” population, followed for a long period of time. MR imaging acquisition and postprocessing, statistical methodology, and use of VLSM analysis are in line with the high quality of previous studies of this team—molecular analysis prevents any significant criticism, with the limit of exploring only somatic point mutations. The strength of the VLSM tool is, compared with a classical “anatomically based” atlas, to identify with a better spatial localization and discrimination the exact region of IDH-mutated glioma cells. Interestingly, *none of the remaining molecular markers was associated with a specific tumor location*, suggesting that EGFR amplification or TERT status was not a driver of initial oncogenic events.

Visualizing this niche paves the way for real “personalized” treatments, such as delivering within this area (curative or preventive) higher doses of radiation or local convection-enhanced or targeted therapy. In addition, with the recent ability to non-invasively quantify D-2-hydroxyglutarate accumulation using sophisticated magnetic resonance spectroscopy techniques,^{9,10} there is a place for “piloting” the dynamic evolution of this specific zone during and after treatment. Because VLSM is a freely available toolbox, these possibilities could be potentially shared within a prospective multicenter trial.

So, what are the potential issues? First, and as for all advanced MR techniques, there will be a need for homogenization and diffusion of image acquisition protocols and postprocessing software. Second, in the RT field, coregistering all of these “maps” and integrating them robustly into our treatment planning

software will be a challenge. Third, these hypotheses can only be tested for IDH-*mutated* gliomas, with clear founder events for IDH-*wild type* gliomas remaining to be uncovered.

Disclaimer

This text is the sole product of the authors and no third party had input or gave support to its writing.

References

1. Weller M, van den Bent M, Tonn JC, et al; European Association for Neuro-Oncology (EANO) Task Force on Gliomas. European Association for Neuro-Oncology (EANO) guideline on the diagnosis and treatment of adult astrocytic and oligodendroglial gliomas. *Lancet Oncol*. 2017;18(6):e315–e329.
2. Niyazi M, Brada M, Chalmers AJ, et al. ESTRO-ACROP guideline “target delineation of glioblastomas”. *Radiother Oncol*. 2016;118(1):35–42.
3. Lim DA, Cha S, Mayo MC, et al. Relationship of glioblastoma multiforme to neural stem cell regions predicts invasive and multifocal tumor phenotype. *Neuro Oncol*. 2007;9(4):424–429.
4. Khalifa J, Tensaouti F, Lusque A, et al. Subventricular zones: new key targets for glioblastoma treatment. *Radiat Oncol*. 2017;12(1):67.
5. Louis DN, Perry A, Reifenberger G, et al. The 2016 World Health Organization classification of tumors of the central nervous system: a summary. *Acta Neuropathol*. 2016;131(6):803–820.
6. Tejada Neyra MA, Neuberger U, Reinhardt A, et al. Voxel-wise radiogenomic mapping of tumor location with key molecular alterations in patients with glioma. *Neuro Oncol*. 2018;20(11):1517–1524.
7. Ellingson BM, Lai A, Harris RJ, et al. Probabilistic radiographic atlas of glioblastoma phenotypes. *AJNR Am J Neuroradiol*. 2013;34(3):533–540.
8. Naeini KM, Pope WB, Cloughesy TF, et al. Identifying the mesenchymal molecular subtype of glioblastoma using quantitative volumetric analysis of anatomic magnetic resonance images. *Neuro Oncol*. 2013;15(5):626–634.
9. Kickingereder P, Andronesi OC. Radiomics, metabolic, and molecular MRI for brain tumors. *Semin Neurol*. 2018;38(1):32–40.
10. Branzoli F, Di Stefano AL, Capelle L, et al. Highly specific determination of IDH status using edited in vivo magnetic resonance spectroscopy. *Neuro Oncol*. 2018;20(7):907–916.